# DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

# HERIOT-WATT UNIVERSITY

## F78AB Actuarial and Financial Mathematics B

**Date: 23 May 2014**

**Time: 09:00 to 11:00**

| Question | Marks |
|---|---|
| 1 | 8 |
| 2 | 6 |
| 3 | 6 |
| 4 | 12 |
| 5 | 10 |
| 6 | 18 |
| Total Marks | 60 |

**Attempt ALL 6 questions.**

**Show all steps in your calculation.**

**Approved electronic calculators may be used.**

***You should have a copy of the***
**Formulae and Tables for Actuarial Examinations**
***on your desk for this exam.***

**1.** Assume that time is measured in years. The annual force of interest $\delta$ depends on time, with

$$\delta(t) = \begin{cases} 0.07 - 0.005t, & \text{for } t \in [0, 4) \\ 0.055, & \text{for } t \in [4, \infty) \end{cases}$$

(a) Calculate the accumulation at time $t = 10$ of £1 000 invested at time $t = 0$.

[3 marks]

(b) Calculate the present value at time $t = 0$ of a continuous payment stream at the rate of $£100e^{0.065t}$ paid from $t = 5$ to $t = 15$.

[5 marks]

**[Total 8 Marks]**

**2.** The current market price of a stock is £10. The stock pays dividends annually. The next dividend of 50p is payable in two months' time. A six-month forward contract is available on the stock for a forward price of £9.50. The risk-free force of interest is 3% per annum.

Show how an investor can make a risk-free profit.

**[Total 6 Marks]**

**3.** (a) In a market,

- the 5-year spot rate is $y_5 = 4.75\%$ per annum effective, and
- the 3-year forward rate from time 2 is $f_{2,3} = 4.25\%$ per annum effective.

Calculate the 2-year spot rate of interest assuming no-arbitrage.

[3 marks]

(b) In another market, today the prices of £1 nominal of a 1-year, a 2-year and a 3-year zero-coupon bond are $P_1 = 0.95$, $P_2 = 0.90$, and $P_3 = 0.85$, respectively.

Also today in the market, the government issues a 3-year fixed-interest bond with coupons of 5% per annum payable annually in arrears, and redeemable at par.

Calculate the issue price of the bond per £100 nominal, stating any assumption that you make about the market.

[3 marks]

**[Total 6 Marks]**

[PLEASE TURN OVER]

**4.** A cash sum of £1 000 is invested in a fund and held for 15 years. The yield on the investment in any year will be 5% with probability 0.2, 7% with probability 0.6 and 9% with probability 0.2, and is independent of the yield in any other year.

(a) Calculate the mean accumulation at the end of 15 years.

[3 marks]

(b) Calculate the standard deviation of the accumulation at the end of 15 years.

[5 marks]

(c) Without carrying out any further calculations, explain how your answers to parts (a) and (b) would change (if at all) if:

(i) the yields had been 4%, 7% and 10% instead of 5%, 7%, and 9% per annum, respectively.

(ii) the investment had been held for 18 years instead of 15 years.

[4 marks]

**[Total 12 Marks]**

[PLEASE TURN OVER]

**5.** An index-linked bond was issued on 18 October 2013. The nominal coupon rate is 2% per annum and coupons are paid every half-year, on 18 April and 18 October each year. The bond is to be redeemed at 100% on 18 October 2023.

The redemption payment and all coupon payments are increased in line with the RPI, using a 3-month time lag. The base value for indexation purposes is the July 2013 value of the RPI.

The last six published values of the RPI are shown in the table below.

| Date | Jul 2013 | Aug 2013 | Sep 2013 | Oct 2013 | Nov 2013 | Dec 2013 |
|---|---|---|---|---|---|---|
| RPI (Date) | 249.7 | 251.0 | 251.9 | 251.9 | 252.1 | 253.4 |

Assume that the inflation rate between December 2013 and October 2023 is a constant 3% per annum effective.

(a) Calculate the coupon payment received on 18 April 2014 by an investor holding £100 nominal of the bond.

[2 marks]

(b) Calculate the redemption payment received on 18 October 2023 by an investor holding £100 nominal of the bond.

[2 marks]

On 18 December 2013, the bond is priced so that a tax-exempt investor gets a money yield of 6.5% per annum.

(c) Calculate the price on 18 December 2013 of £100 nominal of the bond.

[5 marks]

(d) Without doing any calculations, state how the money yield on the bond changes if future inflation is always higher than a constant 3% per annum. Briefly explain your reasoning.

[1 mark]

**[Total 10 Marks]**

[PLEASE TURN OVER]

**6.** (a) A variable annuity, payable annually in arrears for an integer number of years $n$, is such that the payment at the end of year $k$ is $k^2$, for $k = 1, 2, \ldots, n$.

Show that the present value of this annuity can be expressed as

$$\frac{2(Ia)_{\overline{n}|} - a_{\overline{n}|} - n^2 v^{n+1}}{1 - v},$$

under the assumption of a constant interest rate.

[4 marks]

(b) Today a pension fund has a liability to pay £200 000 in exactly one year's time, £210 000 in exactly two years' time and so on, the amount increasing by £10 000 each year to £390 000 at the end of 20 years, this being the last payment.

The fund values these payments using a force of interest of 5% per annum. This is also the force of interest at which the current prices of all bonds are calculated.

The fund owns as assets:

- £$X$ nominal of a zero–coupon bond redeemable at par in 25 years, and
- £$Y$ nominal of a fixed interest bond redeemable at par in 12 years' time which pays a coupon of 6% per annum annually in arrears.

Minor discrepancies due to rounding errors in your answers to the questions below will not be penalised.

(i) Show that the present value and the duration of the liabilities are £3 435 425 and 10.000 years, respectively.

[6 marks]

(ii) Show that an expression for the present value of the assets is:

$$0.28650X + 1.07681Y.$$

[2 marks]

(iii) Show that an expression for the duration of the assets is:

$$\frac{7.16250X + 9.70502Y}{0.28650X + 1.07681Y}.$$

[3 marks]

(iv) Hence calculate the values of $X$ and $Y$ such that the present value and duration of the assets are equal to those of the liabilities.

[3 marks]

**[Total 18 Marks]**

[END OF PAPER]